Quality and Compliance Manual

Quality Department rev. 7

1. Scope and Certification

This manual defines the quality management system of Halvorsen Industrial Systems for the design, manufacture and servicing of industrial pumps and controllers.

The quality management system is certified to ISO 9001:2015 under certificate number QMS-48213. The certification body is Nordic Assurance Services.

The Quality Director is the management representative for the QMS. A management review of the QMS is held every 6 months.

The quality manual itself is reviewed annually by the Document Control team.

2. Calibration of Measuring Equipment

All inspection, measuring and test equipment is uniquely identified and recorded in the calibration register.

Torque wrenches are calibrated every 6 months to a tolerance of plus or minus 2 percent. Pressure gauges are calibrated every 12 months.

Coordinate measuring machines are verified against reference artefacts every 90 days. Equipment found out of tolerance is tagged with a red QUARANTINE label and removed from use.

Calibration records are retained for 10 years.

3. Nonconformance and Corrective Action

Any product, process or supplier nonconformance is recorded in the quality system and assigned an owner.

Nonconformance reports (NCRs) must be closed within 21 days of being raised. Root cause analysis for corrective actions uses the 8D method.

The effectiveness of each corrective action is verified 90 days after implementation. Customer complaints must be acknowledged within 2 working days.

A Material Review Board meets every Tuesday to disposition nonconforming material.

4. Internal and Supplier Audits

Audits are planned on a risk basis and carried out by trained auditors who are independent of the area audited.

Every process in the QMS is internally audited at least once every 12 months. Critical suppliers are audited on site every 24 months.

Internal auditors must complete a 3-day lead auditor course before conducting audits. Major audit findings must have a corrective action plan submitted within 10 days.

Suppliers with a quality score below 70 are placed on probation.

5. Product Release and Traceability

Finished products are released only after all inspection and test steps defined in the control plan are complete.

Every pump undergoes a hydrostatic pressure test at 1.5 times its rated pressure before release. Pump serial numbers follow the format HX-YYWW-NNNN.

Traceability records for safety-critical components are retained for 15 years. Final inspection sampling follows ISO 2859-1 with an AQL of 0.65.

The Quality Engineer signs the certificate of conformity for each shipment.